

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Toolbox	1	hr

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9702/31

October/November 2018

2 hours

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be used only if it becomes necessary to do so.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **11** printed pages and **1** blank page.

You may not need to use all of the materials provided.

1 In this experiment, you will investigate the motion of a wooden strip.

(a) • Set up the apparatus as shown in Fig. 1.1.

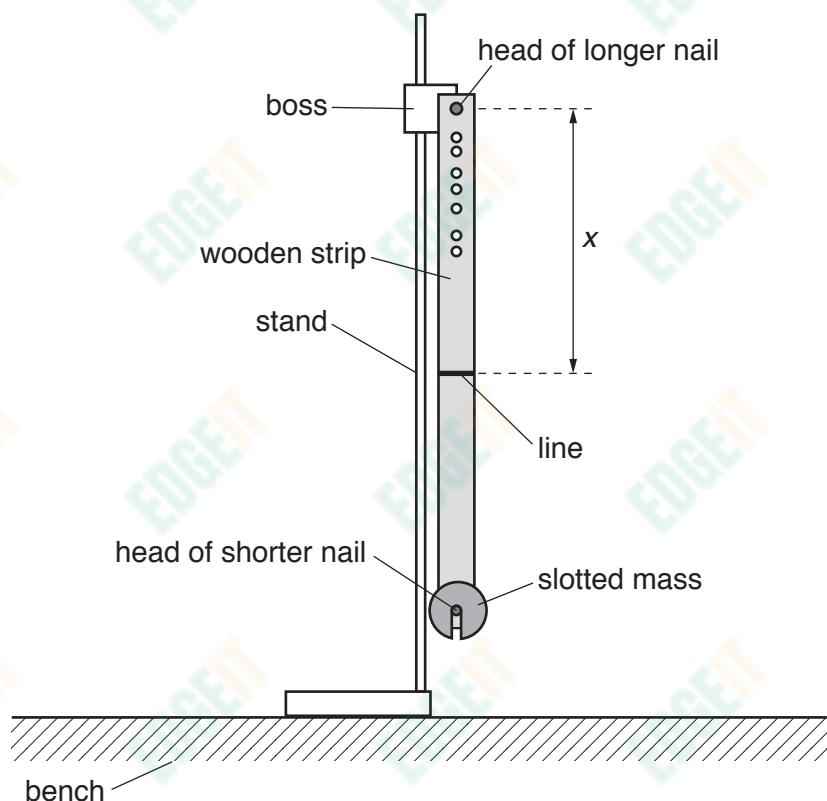


Fig. 1.1

- The longer nail should be held in the boss and be placed through the top hole in the strip.
- The shorter nail should be placed through the bottom hole in the strip to support the slotted mass.
- The distance between the longer nail and the line on the strip is x , as shown in Fig. 1.1.

Measure and record x .

$x = \dots\dots\dots$ [1]

- (b) • Pull the bottom of the strip through a short distance, as shown in Fig. 1.2.

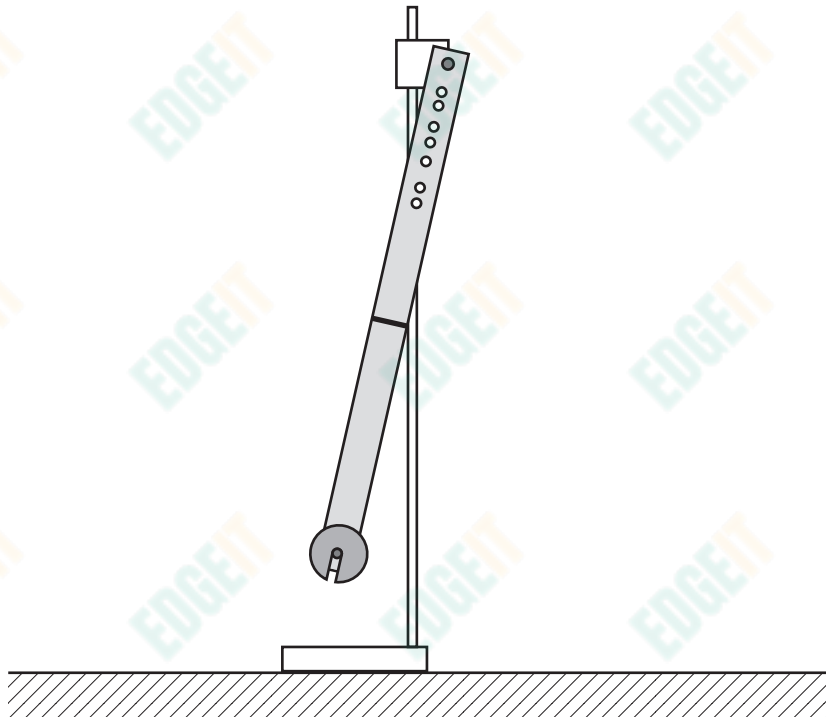


Fig. 1.2

- Release the strip. The strip will oscillate.
- Determine the period T of these oscillations.

$T = \dots\dots\dots$ [1]

- (c) Vary x by placing the longer nail through different holes in the strip. Measure and record x and T . Repeat until you have six sets of values.

Record your results in a table. Include values of T^2x and x^2 in your table.

[10]

- (d) (i) Plot a graph of T^2x on the y -axis against x^2 on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

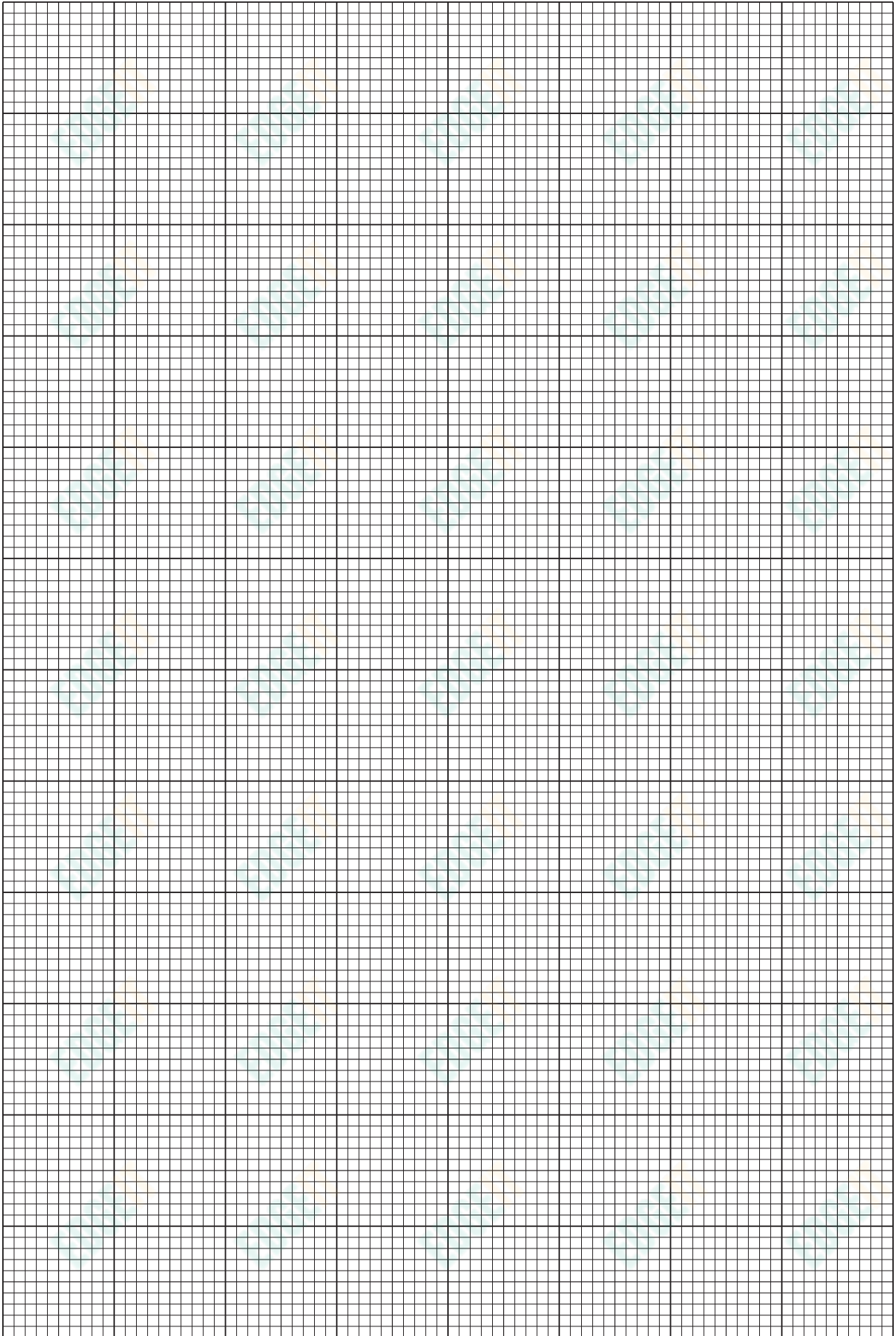
[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]



- (e) It is suggested that the quantities T and x are related by the equation

$$T^2x = Ax^2 + B$$

where A and B are constants.

Using your answers in (d)(iii), determine the values of A and B .
Give appropriate units.

$A =$

$B =$ [2]

[Total: 20]

You may not need to use all of the materials provided.

2 In this experiment, you will investigate the movement of a chain of paper clips.

(a) (i) You have been provided with a chain of paper clips.

Measure and record the mass M of the chain.

$M =$ [1]

(ii) Using your answer in **(a)(i)**, calculate the mass m of **one** paper clip.

$m =$ [1]

(iii) Justify the number of significant figures that you have given for your value of m .

.....

 [1]

(b) (i) You have been provided with two identical spheres of modelling clay.

Measure and record the mass S of one of the spheres.

$S =$ [1]

(ii) Calculate c where

$$c = \frac{S}{m}.$$

$c =$ [1]

- (c) (i) • Place the wooden strip on the bench so that one end of the strip overhangs the bench, as shown in Fig. 2.1.

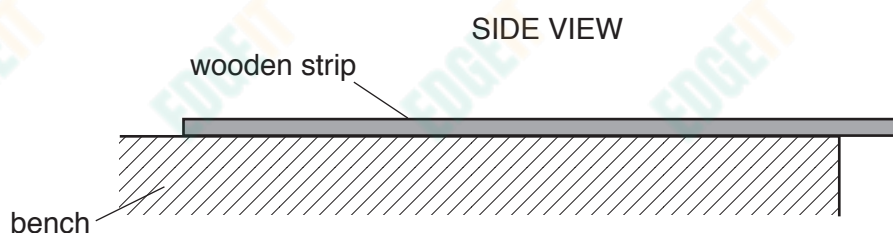


Fig. 2.1

- Place the chain on the wooden strip and place one of the spheres on the paper clip at the end of the chain, as shown in Fig. 2.2. The sphere must **not** touch the wooden strip.

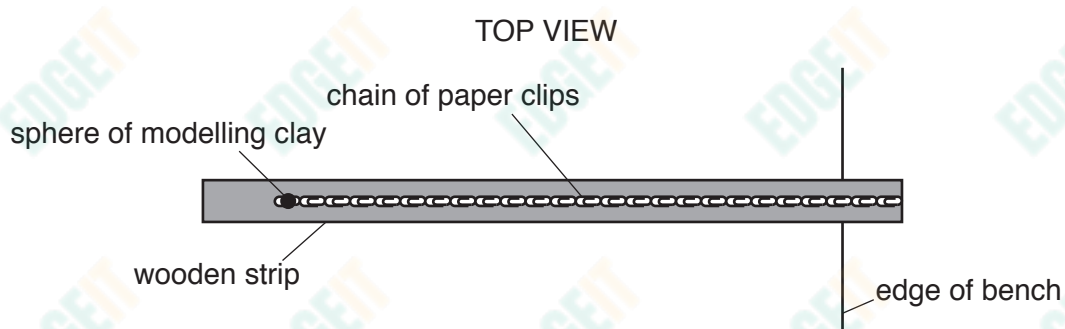


Fig. 2.2

- Gradually move the paper clips off the end of the wooden strip so that they hang down, as shown in Fig. 2.3.

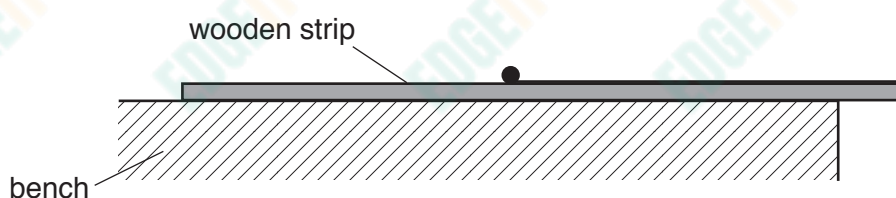


Fig. 2.3

- When the chain starts to slip, count and record the number p of paper clips hanging off the end of the wooden strip and the number q of paper clips **remaining** on the strip.

$p =$

$q =$

[2]

- (ii) Estimate the percentage uncertainty in your value of p .

percentage uncertainty = [1]

- (d) • Replace the chain on the wooden strip and place both spheres on the paper clips at the end of the strip, as shown in Fig. 2.4.

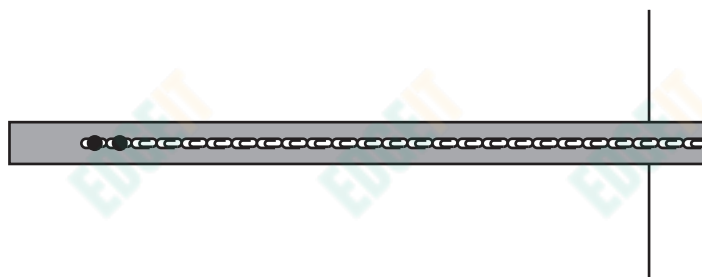


Fig. 2.4

- Determine p and q .

$p =$

$q =$

[2]

- (e) It is suggested that the relationship between p , q and c is

$$p = k(q + nc)$$

where n is the number of spheres on the chain and k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

- (ii) Explain whether your results support the suggested relationship.

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..... [1]

- (f) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.
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2.
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3.
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4.
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[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.
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2.
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3.
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4.
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[4]

[Total: 20]

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